

Introduction

Neurodegenerative diseases such as Alzheimer’s and Parkinson’s disease are characterized by progressive neural damage resulting from protein misfolding and aggregation. Heat shock proteins (HSPs) are molecular chaperons that maintain proteostasis by preventing aggregation under cellular stress. Reduced HSP activity leads to an increase in protein damage.

Natural polyphenolic compounds have antioxidant properties that enhance cellular stress resistance. Curcumin has shown to moderate protein aggregation pathways. However, dose-dependent effects of curcumin on heat-induced protein damage remain inconclusive.

Hypothesis

If *C. elegans* are treated with 10 µM, 25 µM, and 50 µM curcumin prior to heat stress, then worms treated with 25 µM will show the greatest enhancement of heat shock protein proteostasis, while lower doses may be insufficient and higher doses may produce diminishing effects.

Relevance

This study aims to identify the optimal curcumin dosage to enhance heat shock protein function. The model organism *Caerohabtitis elegans* is used to study proteostasis due to its heat shock response and transparent neurological system, which allows for fluorescence-based imaging.

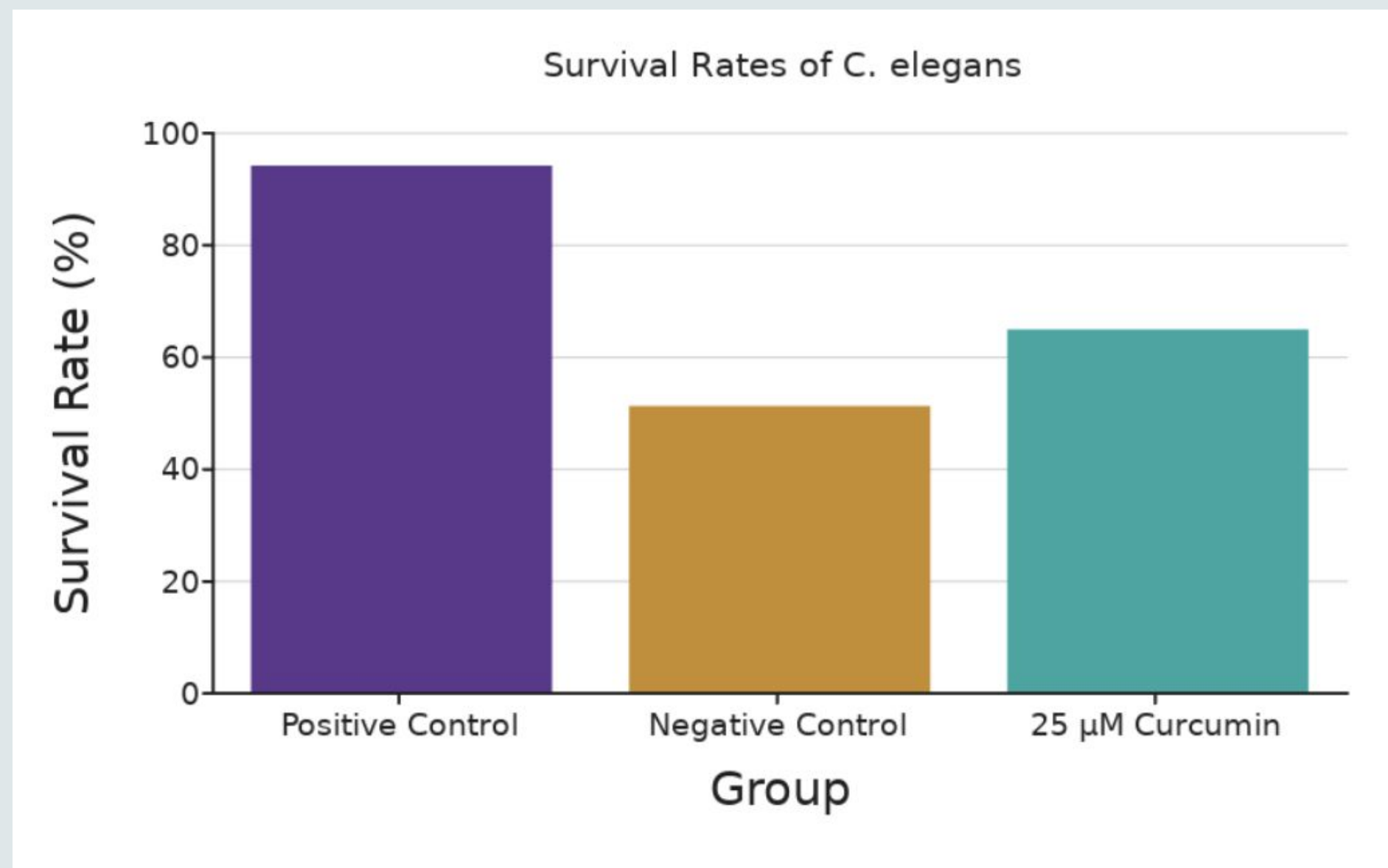


Figure 1. Graph made by researcher using Data Classroom of survival rates of *C. elegans* across all treatments

Dose-Dependent Effects of Curcumin on Heat Shock Protein Function and Heat Stress Resistance on *Caenorhabditis elegans*

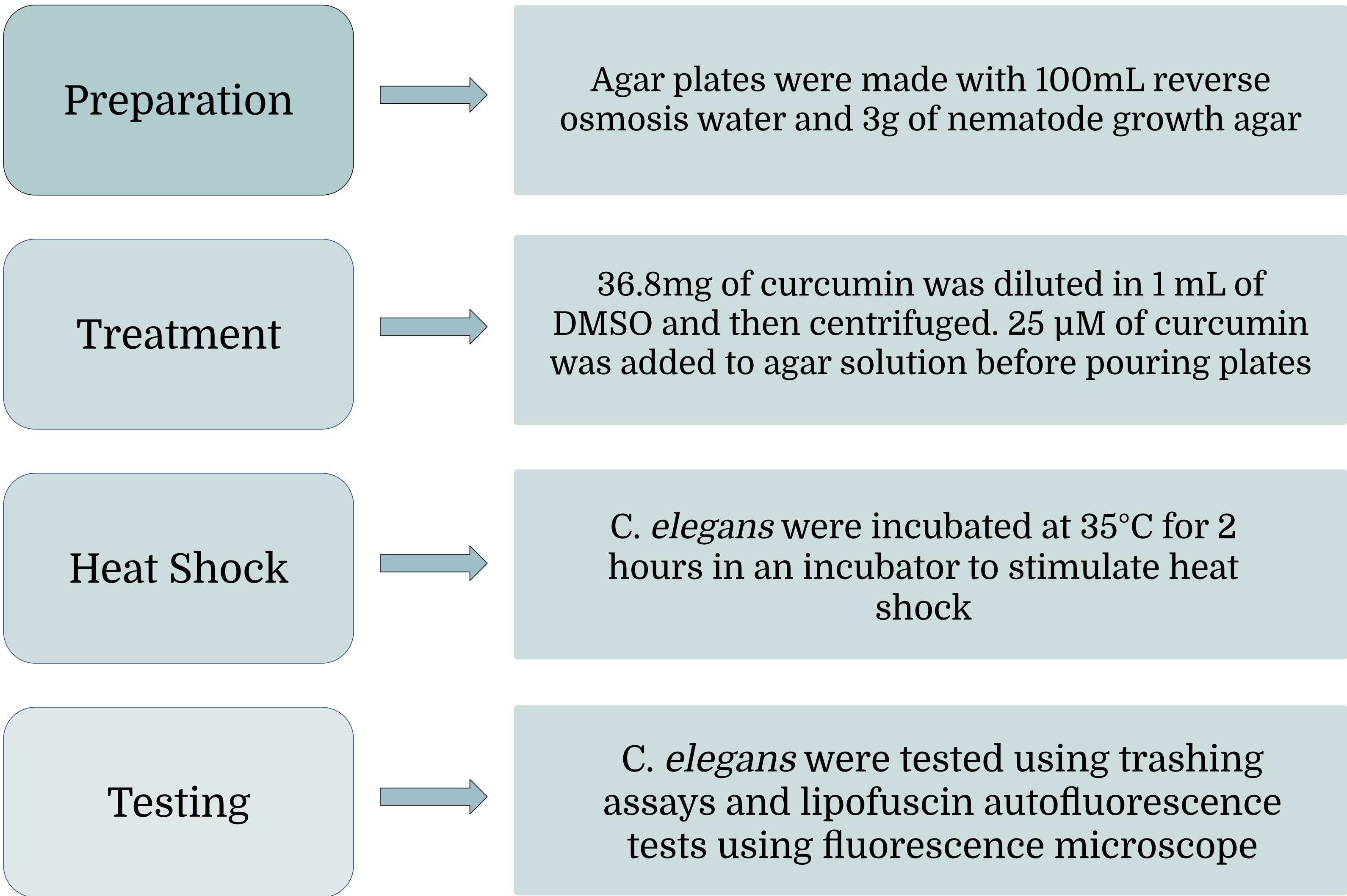


Figure 2. Photo taken by researcher of *C. elegans* being chunked onto treatment plates, 2026



Figure 3. *C. elegans* Photo taken by researcher, 2026

Results - Fluorescence

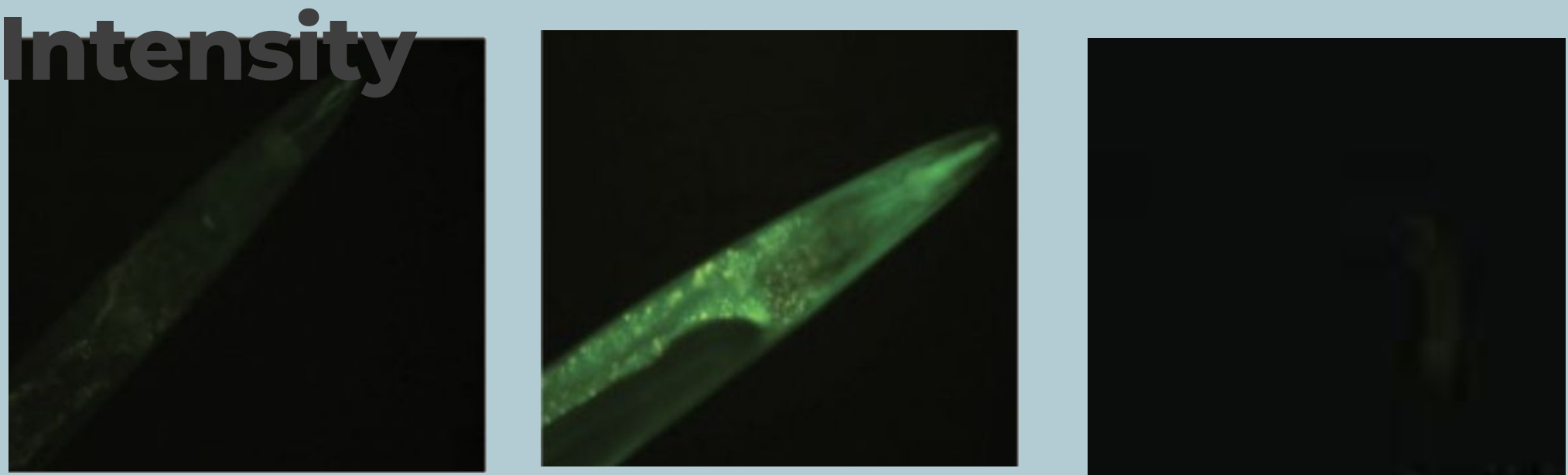


Figure 5. Figure 6. Figure 7.

Figure 5. Fluorescence imaging of *C. elegans*. From Research Gate [Photograph], by Jatinder Sangha, 2012
Figure 6. Fluorescence imaging of *C. elegans* with Thermal Stress From Research Gate [Photograph], by Jatinder Sangha, 2012
Figure 7. *C. elegans* treated with 25µM of Curcumin Fluorescence image taken by researcher, 2026

Data Analysis

Pair	P-value	Interpretation of P
25 µM - Positive Control	<0.01	Extremely strong confidence that the groups are different
25 µM - Negative Control	<0.01	Extremely strong confidence that the groups are different
Positive Control - Negative Control	<0.01	Extremely strong confidence that the groups are different

Figure 8. Table made by researcher of results from a Kruskal-Wallis test and made using Data Classroom.

Conclusion/Future Research

Curcumin enhances protein function of *C. elegans* under heat shock conditions, as shown by survival and thrashing assays. These findings support the potential role of curcumin in modulation to cellular response pathways.

Future research is needed to determine the potential of curcumin in enhancing the role of heat shock protein expression and dose-dependent responses. Further research can investigate curcumin’s protective effects in terms of neurodegenerative diseases.

References

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https://doi.org/10.1007/978-0-387-39975-1_15
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Whyte, B. (2023, July 20). *Misfolded proteins and neurodegenerative diseases* | BMG LABTECH. www.bmglabtech.com. <https://www.bmglabtech.com/en/blog/misfolded-proteins-and-neurodegenerative-diseases>

Results - Thrashing Assay

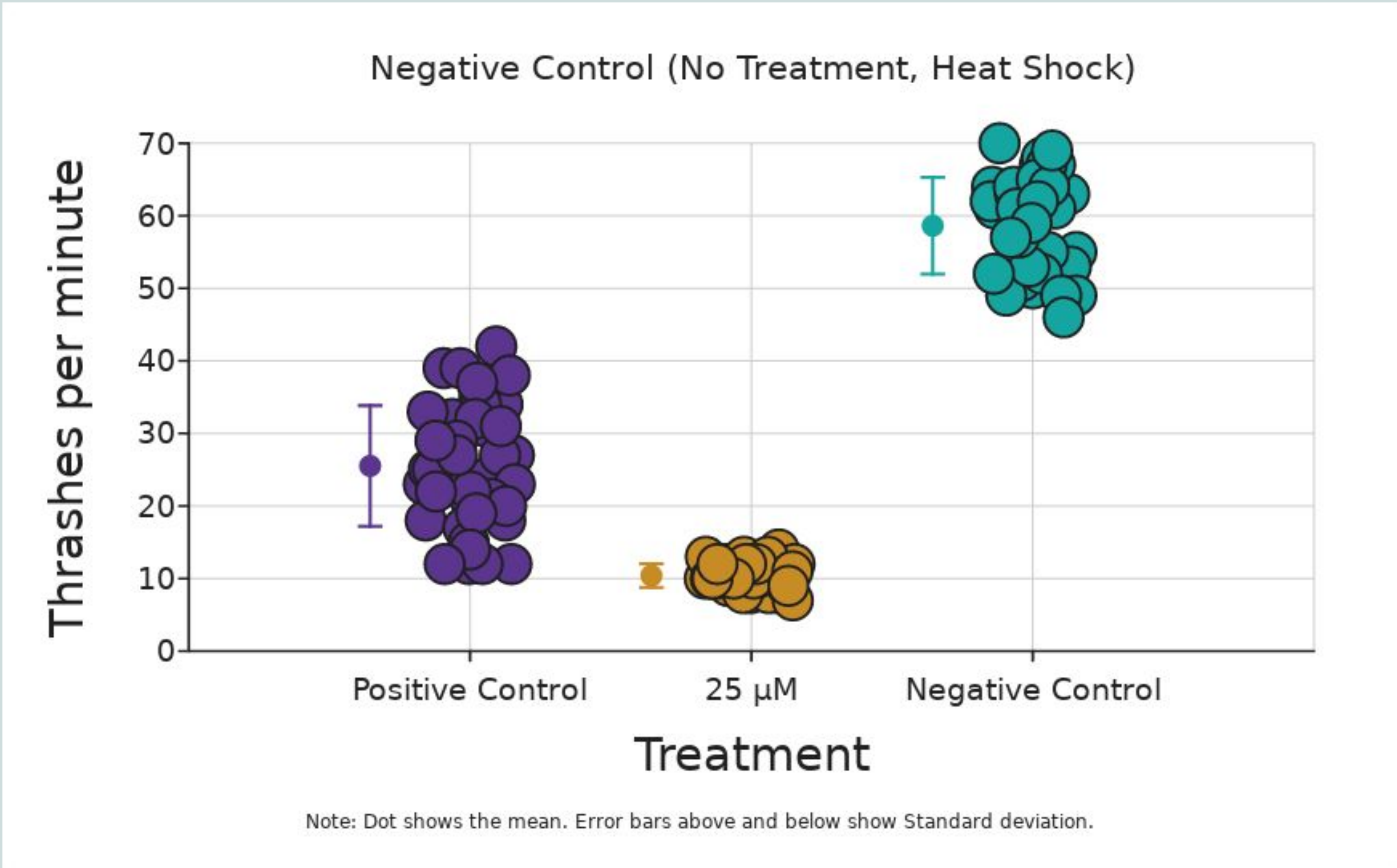


Figure 4. Graph made by researcher using Data Classroom of thrashes per minute for each experimental treatment